

Introduction:

In this project I performed EDA to explore and understand the Ecommerce Sales Database. The database has two dimension tables and one fact table. I analyzed dimensions, measures and dates to understand overall businesses. I also highlighted the missing values in the database.

Database Information

	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	TABLE_TYPE
1	DataWarehouseAnalytics	gold	customers_report	VIEW
2	DataWarehouseAnalytics	gold	dim_customers	BASE TABLE
3	DataWarehouseAnalytics	gold	dim_products	BASE TABLE
4	DataWarehouseAnalytics	gold	fact_sales	BASE TABLE
5	DataWarehouseAnalytics	gold	products_report	VIEW

Summary:

Database Name = DataWarehouseAnalytics

Schema Name = gold

No_of_tables = 3

No_of_views = 2

Exploratory Data Analysis

1. Countries our customers come from

	country	no_of_customers
1	Australia	3591
2	Canada	1571
3	France	1810
4	Germany	1780
5	n/a	337
6	United Kingdom	1913
7	United States	7482

2. All Categories

	category
1	NULL
2	Accessories
3	Bikes
4	Clothing
5	Components

3. Count of Sub-Categories by Categories

	category	count_of_sub_categories
1	NULL	0
2	Accessories	12
3	Bikes	3
4	Clothing	8
5	Components	13

4. Products with missing categories and sub-categories

	product_name	category	subcategory
1	LL Mountain Pedal	NULL	NULL
2	ML Mountain Pedal	NULL	NULL
3	HL Mountain Pedal	NULL	NULL
4	LL Road Pedal	NULL	NULL
5	ML Road Pedal	NULL	NULL
6	HL Road Pedal	NULL	NULL
7	Touring Pedal	NULL	NULL

5. First order date and last order date

Only 3 days sales data in 2010 and 28 days in 2014

Around 37 months of Sales Data is Present

	first_order_date	last_order_date
1	2010-12-29	2014-01-28

6. Big Numbers Exploration

	measure_name	measure_value
1	Total Sales	29356250
2	Total Quantity	60423
3	Avg Selling Price	486
4	Total Orders	27659
5	Total Products	295
6	Total Customers	18484

7. Customers Distribution By gender

	gender	no_of_customers
1	Female	9128
2	Male	9341
3	n/a	15

8. Products Distribution by Products

	category	no_of_products
1	NULL	7
2	Accessories	29
3	Bikes	97
4	Clothing	35
5	Components	127

9. Avg Cost in each category

	category	avg_cost
1	NULL	28
2	Accessories	13
3	Bikes	949
4	Clothing	24
5	Components	264

10. Revenue By Category

	category	total_sales
1	Clothing	339716
2	Bikes	28316272
3	Accessories	700262

11. Top 10 customers By revenue

	customer_number	full_name	total_revenue
1	AW00012301	Nichole Nara	13294
2	AW00012132	Kaitlyn Henderson	13294
3	AW00012308	Margaret He	13268
4	AW00012131	Randall Dominguez	13265
5	AW00012300	Adriana Gonzalez	13242
6	AW00012321	Rosa Hu	13215
7	AW00012124	Brandi Gill	13195
8	AW00012307	Brad She	13172
9	AW00012296	Francisco Sara	13164
10	AW00011433	Maurice Shan	12914

12. Distribution of Sold items across countries

	country	items_sold
1	n/a	871
2	Germany	5626
3	United States	20481
4	Australia	13346
5	United Kingdom	6910
6	Canada	7630
7	France	5559

13. Top 5 products by Revenue

	product_name	total_revenue
1	Mountain-200 Black- 46	1373454
2	Mountain-200 Black- 42	1363128
3	Mountain-200 Silver- 38	1339394
4	Mountain-200 Silver- 46	1301029
5	Mountain-200 Black- 38	1294854

14. Bottom 5 products By Revenue

	product_name	total_revenue
1	Racing Socks- L	2430
2	Racing Socks- M	2682
3	Patch Kit/8 Patches	6382
4	Bike Wash - Dissolver	7272
5	Touring Tire Tube	7440